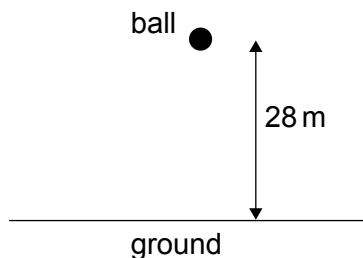


PAPER 2 / QUESTION 18

26 April 2024 / TZ2 (file label) / HL Paper 2 / original question 1 / 2224-6708

18. A ball of mass 2.7 g is released from rest from a height of 28 m above horizontal ground.



- (a) Show that in the absence of air resistance the ball impacts the ground with a speed of about 23 ms^{-1} .

[1]

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- (b) An air resistance force F acts on the ball. F can be modeled by $F = kv^2$ where v is the speed and k is a constant.

- (i) Determine the unit of k in terms of fundamental units.

[2]

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- (ii) Describe how the ball reaches terminal speed.

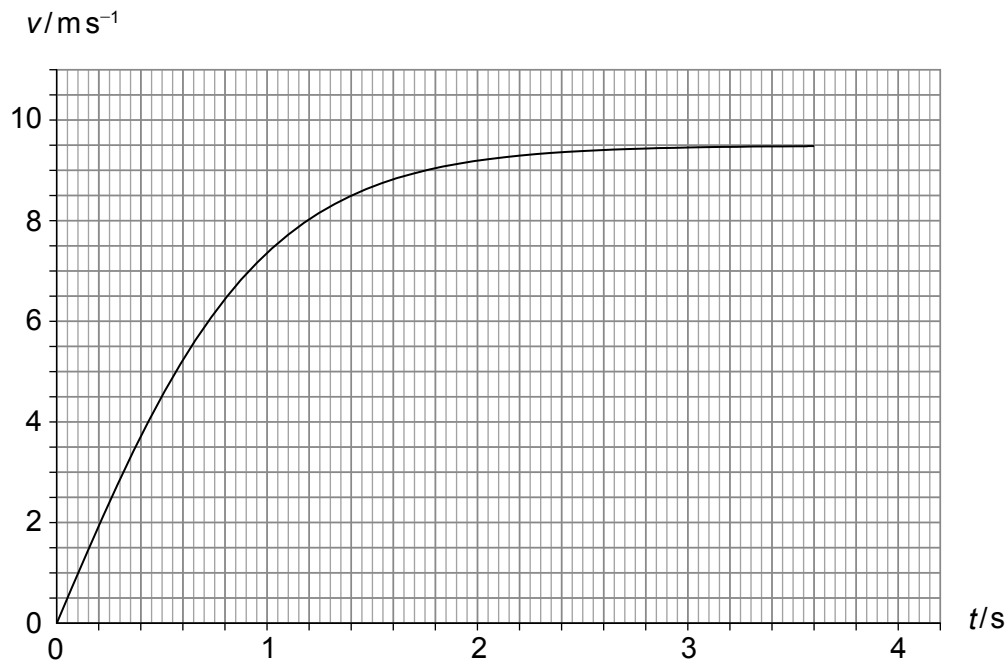
[2]

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(Question 18 continued)

- (c) The graph shows the variation with time t of the speed v of the ball from the instant it is released until it impacts the ground.



- (i) State the value of the area under the curve. [1]

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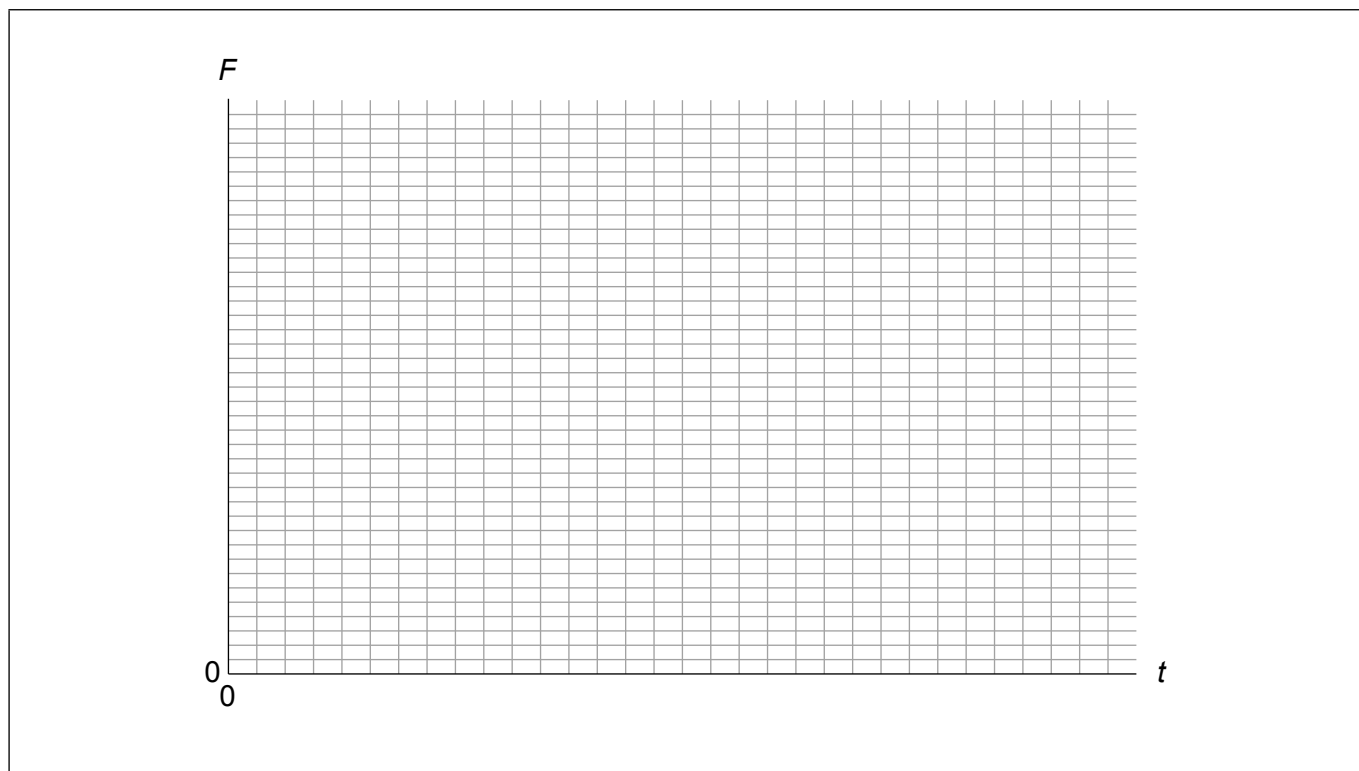
- (ii) Determine k . [2]

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(Question 18 continued)

- (iii) On the axes below, draw a graph to show the variation of the magnitude of the **resultant** force, F , on the ball with time t . No numbers are required on the axes. [1]



- (iv) Calculate the average power dissipated by the air resistance force. [3]

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- (d) The ball rebounds from the ground with speed 7.8 m s^{-1} . The ball is in contact with the ground for a time T . The average **resultant** force on the ball during this time is 1.1 N .

Determine T . [2]

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